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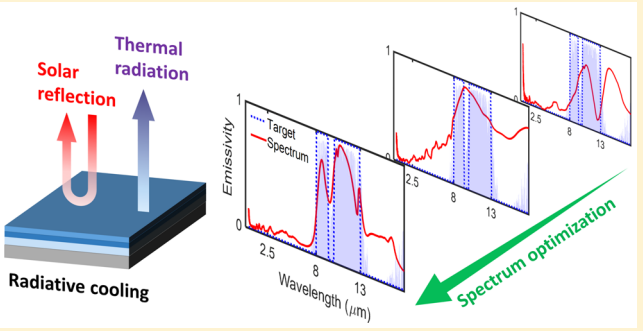
Optimization of Multilayer Optical Films with a Memetic Algorithm and Mixed Integer Programming

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Supporting Information

ABSTRACT: Multilayer optical films have been extensively used in optical technology, but the design of multilayer structures for broadband applications is often challenging due to the need to incorporate material dispersion. Here, we present an implementation of a memetic algorithm based on mixed integer programming, which is especially suited for practical broadband optimization of layered thin-film materials. In our implementation, the optimization variables consist of a list of discrete variables that represents different dielectric materials, along with a list of continuous variables that represents the thicknesses of each material. As a set of concrete demonstrations, we optimize the spectra of a radiative cooling device and an incandescent light bulb filter. The resulting structures from the optimization can, by using more materials, achieve better performance than their counterparts in the literature while using fewer numbers of material layers.



KEYWORDS: *evolutionary algorithm, photonics optimization, spectral design, radiative cooling, thermal emitter filter*

Multilayer optical thin films have been widely used to achieve specific optical transmission and reflection spectra for applications in many important areas of optical technology in the last century.[1−3] In recent years, there has also been significant interest in designing multilayer optical structures for energy and thermal applications.[4−15] Notable examples include efforts to improve the efficiency of solar and thermophotovoltaic cells,[4,8−10] to achieve passive radiative cooling,[7,11] to attain broadband spectral filtering,[12,13] and to enhance the efficiency of thermal and optical emitters.[14,15] A common characteristic for energy and thermal applications is the need to control optical properties over a broad range of wavelengths. For example, in the passive radiative cooling application, one needs to simultaneously achieve particular spectral properties across both solar and thermal wavelength ranges. For such solar and energy applications, a systematic and robust multilayer design technique for broadband optical spectrum engineering is thus highly desirable.

There exist many methods of designing multilayer structures. One can certainly rely on physical intuition,[2,12,14,16] but doing so tends to explore only a tiny subspace of the entire allowed design phase space. Therefore, a wide range of systematic approaches in designing optical multilayer structures have been explored. A few prominent examples include the jump method,[17] spiral search method,[18] particle swarm optimization,[19] needle optimization,[1,20−22] and evolutionary algorithms.[23−25] When implementing these methods, the optimization parameters are usually chosen to be the thicknesses and refractive indices of each layer. As it turns out, the choice of refractive indices as optimization parameters strongly limits the practical usefulness of these algorithms in applications that require optimization over broad bandwidths. In practice, one typically fabricates multilayer structures with a discrete set of materials. Over a broad bandwidth, the refractive indices of all materials vary as a function of wavelength. In the usual implementations, incorporating such material dispersion is difficult since with material dispersion, a given material does not have a single corresponding refractive index. As an indication of this difficulty, many previous works on optical multilayer design in fact assume nondispersive materials. In order to design a multilayer structure for broadband energy and thermal applications, it is essential to develop a practical algorithm that starts from a given list of materials with their realistic dispersions.

In this work, we propose and test an implementation of a memetic algorithm, the combination of an evolutionary algorithm and local optimization method,[26,27] to design multilayer devices starting from a list of preselected materials. In contrast to previous work in this space, we implement the optimization algorithm as a mixed-integer programming (MIP) problem, where the optimization variables consist of a list of discrete variables that represent different optical materials, along with a list of continuous variables that represent the thicknesses of each material. In the following sections, we present the details of this algorithm's formulation, and as a demonstration, we apply the memetic algorithm to the optimization of a passive radiative cooling device[7] and

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incandescent light bulb filter.[14] We show that the structures produced from this algorithm have better performance than those reported in literature while using fewer numbers of layers, demonstrating that our memetic algorithm is especially suitable for broadband thin-film optimization.

MEMETIC ALGORITHM IMPLEMENTATION

To start, we outline how we implement the memetic algorithm for thin-film optimization. First, we suppose that the incident and substrate media have refractive indices $n_{in}(\lambda)$ and $n_{out}(\lambda)$, respectively, where λ is the wavelength. To design a multilayer structure, we specify a target reflection spectrum denoted as $R^*(\lambda, \theta)$, where θ is the incident angle. Given a structure that consists of K-layers of materials with refractive indices $\mathbf{n}(\lambda) = [n_1(\lambda), ..., n_K(\lambda)]^T$ and thicknesses $\mathbf{d} = [d_1, ..., d_K]^T$, the reflection viewed from the incident medium, $R(\lambda, \theta; \mathbf{n}, \mathbf{d})$, can be efficiently calculated using the impedance method as described in ref 28. To find an optimum thin film structure, we wish to minimize the residual between the reflection spectrum of a given structure and the target spectrum as described by the merit function $F(\mathbf{n}, \mathbf{d})$:

$$F(\mathbf{n}, \mathbf{d}) = \sum_{\lambda,\theta} W(\lambda)(R(\theta, \lambda; \mathbf{n}, \mathbf{d}) - R^*(\theta, \lambda))^2 \quad (1)$$

where $W(\lambda)$ describes the weight associated with each wavelength.

For broadband optimizations, where material refractive indices can vary greatly among wavelengths of interest, it is necessary to accurately account for material dispersion. In many practical situations, one chooses to fabricate a thin film structure using only a discrete set of materials. To account for this, instead of treating the list of refractive indices, $\mathbf{n}(\lambda)$, as continuous variables, we treat the material composition as a list of integers that varies from 1 to M denoted as $\tilde{\mathbf{n}}$. Each of the entries in $\tilde{\mathbf{n}}$ represents a material chosen from a list of M realistic materials. Given a desired spectrum $R^*(\lambda, \theta)$, the optimization problem can be formulated as the following mixed-integer problem:

$$[\tilde{\mathbf{n}}^*, \mathbf{d}^*] = \underset{\tilde{\mathbf{n}}\in\mathbb{Z}^K, \mathbf{d}\in\mathbb{R}^K}{\operatorname{argmin}} F(\tilde{\mathbf{n}}, \mathbf{d}) \quad (2)$$

where $\tilde{\mathbf{n}}^*$ and $\mathbf{d}^*$ represent the optimal material combinations and layer thicknesses, respectively.

Having defined the optimization problem in eq 2, we now outline the use of the memetic algorithm in finding the optimal multilayer structure. The memetic algorithm belongs to the class of evolutionary algorithms, which adapt the biological rule of survival-of-the-fittest to conduct optimization searches that are theoretically and empirically robust in complex spaces.[26,27] In particular, evolutionary algorithms are not constrained by the domain of the input parameters, and therefore they are especially suited for mixed-integer optimization problems such as the one described in eq 2. Our implementation of the memetic evolutionary algorithm consists of the following operations:

1. Generating a Population. We first randomly generate N number of K-layered structures as a set that we call "population": $P^{(0)} = \{[\tilde{\mathbf{n}}^{(1)}, \mathbf{d}^{(1)}], [\tilde{\mathbf{n}}^{(2)}, \mathbf{d}^{(2)}], ..., [\tilde{\mathbf{n}}^{(N)}, \mathbf{d}^{(N)}]\}$. Each structure $[\tilde{\mathbf{n}}^{(i)}, \mathbf{d}^{(i)}]$ is called an "individual". The superscript in the population set P denotes the "generation", which is a counter variable that captures how many times the population has gone through a cycle of steps 2 through 5.

2. Crossover. We mimic the crossover operation to the way that chromosomes exchange genetic information. We group individuals into random pairs $\{[\tilde{\mathbf{n}}^{(i)}, \mathbf{d}^{(i)}], [\tilde{\mathbf{n}}^{(j)}, \mathbf{d}^{(j)}]\}$, which are called "parents". For each set of parents, we randomly select an integer k between 2 and $K - 1$, which denotes the delimiter that separates the parents into their upper sections and lower sections. The upper section of one parent is combined with the lower section of the other, and vice versa, which produces two new individuals that are called "children". Along with the parents, the population size grows to $2N$.

3. Mutation. In every generation, each individual has a small probability μ of experiencing mutation, where we randomly vary the material and thickness of a layer in an individual structure that is randomly chosen. This is to mimic the mutation during biological evolution and allows for the exploration of other search regions.

4. Reselection of Population. We first evaluate eq 1 for each individual and sort the individuals according to their merit functions. We then reselect the next generation of population at a heuristic reselection rate r between 0.8 and 0.99, such that rN individuals are randomly drawn from the top 50th percentile of the current population, and $(1 - r)N$ individuals are randomly drawn from the bottom 50th percentile of the current population. In doing so, the population size is kept at N at the end of each generation. By selecting from both the fit and the unfit portions of the population, we ensure that the population remains sufficiently diverse, which helps in comprehensively exploring the search space.

5. Refinement of Elite Individuals. This step is carried out for every G generations. In this step, we use the quasi-Newton method to perform a local optimization on the thicknesses of the layers in the top P best individuals while keeping the material choice of each layer fixed. Such local optimization in combination with evolutionary algorithm is known to speed up convergence.[27] On the other hand, the process of local optimization is much more computationally demanding than other steps, and therefore, we only perform this refinement to a subset of the population periodically.

Steps 2 through 5 are performed until convergence is reached, which we define as the case when the population's average merit function is within 2% difference compared to the population's minimum merit function. When this occurs, the majority of the individuals will have identical properties, and their merit functions will have arrived at a local minimum that is robust to small fluctuations. During subsequent perturbations to the individuals through, for instance, mutation, the resulting children will likely have larger merit function values and will likely be eliminated through the population reselection step. Therefore, when the population becomes sufficiently uniform, there is negligible probability for further improvement of the population.

It is important to note that when setting up the optimizations for a K-layer structure, it is possible for the algorithm to remove layers during the "crossover", "mutation", and "refinement of elite individuals" steps. This can happen when two identical materials are placed in adjacent layers, or when a particular layer's thickness is set to zero. Thus, the optimized multilayer structure will always have fewer than or equal to K layers, which complies with realistic fabrication constraints.

With the above memetic algorithm formulation, we now apply this method for the optimization of broadband optical spectra. Below, we use the optimization of a radiative cooling device and a filter for an incandescent light bulb as two

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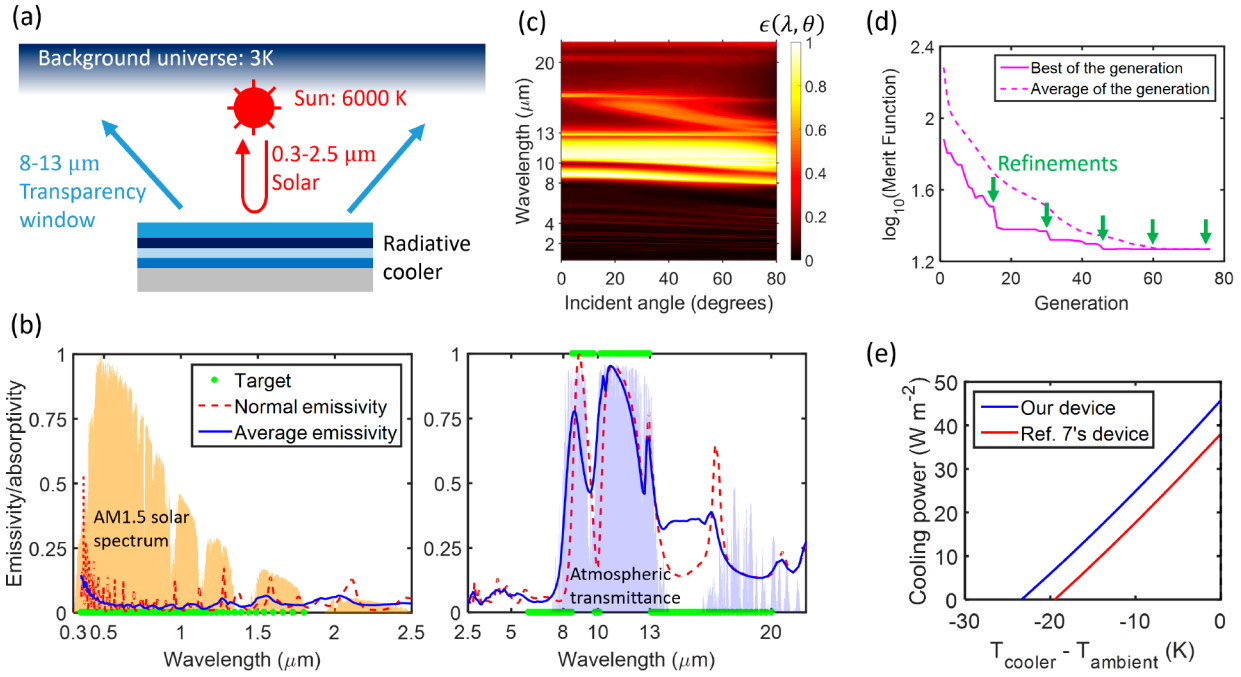


Figure 1. (a) Schematic and operating principles of the radiative cooler. (b) The emissivity/absorptivity spectrum of the designed radiative cooler relative to the solar emission spectrum and the atmosphere transmission window. The green dots are the target emissivity spectrum in the normal direction. The red dotted line and the blue solid line are the normal and angle-averaged emissivity spectrum of the structure in Table 1, respectively. (c) Angle-dependence of the emissivity spectrum. (d) Progression of the radiative cooling device optimization merit function for the best and average of the structures in each generation. The green arrows indicate the generations where the refinement steps took place. (e) Comparison of the temperature dependence of cooling power between our optimized radiative cooling device compared to the radiative cooling device in ref 7.

examples. In particular, we show that our algorithm can effectively select from a list of many materials and produce structures that have superior performance and use fewer layers of materials as compared to their counterparts in the literature.

OPTIMIZATION OF A RADIATIVE COOLING DEVICE

We first demonstrate the application of the memetic algorithm to optimize a radiative cooling device that can operate under direct sunlight. Cooling is a major end-use of energy globally that drives the demand for electricity. For instance, in the United States, powering air conditioning and refrigeration systems accounts for approximately 15 percent of energy used by buildings.[7] Therefore, a passive cooling device without any external energy input can have a significant role in reducing global energy usage. Recently, passive radiative cooling under direct sunlight was demonstrated in ref 7, where a multilayer structure consisting of seven alternating layers of HfO_2 and SiO_2 was placed above a silver reflector, as schematically shown in Figure 1a. The multilayer structure was designed to both be highly reflective over the solar spectrum and highly emissive in the wavelength range between 8 and 13 μm, where the atmosphere has a high transparency window. In the absence of parasitic heating from the environment, and using the atmospheric condition assumed in the theoretical calculations of ref 7, we computed that the structure reported in ref 7 can attain a temperature that is 19.5 K below an ambient temperature of 293 K, even under direct exposure to sunlight.

There are two inter-related metrics by which one can evaluate the performance of radiative cooling device: the net cooling power and the minimum temperature it can attain (its steady-state temperature). On both metrics, the performance of the multilayer structure in ref 7 can be improved. First, the emissivity of the structure in the atmosphere transparency window between 8 and 13 μm can be further increased for more cooling power. Second, within the atmospheric transparency window, there is a significant dip in the atmospheric transmittance between 9.3 and 10 μm due to ozone absorption. Thus, within this wavelength range, there is significant downward radiation from the atmosphere. An ideal radiative cooling device should have low absorptivity/emissivity within this ozone absorption region in order to achieve a lower steady-state temperature.

Here, we employ the memetic algorithm to further optimize the performance of the passive radiative cooling device according to the considerations described above. Compared with ref 7, we choose to have a more diverse set of materials to design a spectrum more favorable for cooling performance. We aim to achieve an objective spectrum that has minimal absorptivity over solar wavelengths, high emissivity in the atmosphere transparency window, and suppressed emissivity between 9.3 and 10 μm. As was done in ref 7, the superstrate and substrate of the cooling device are chosen as air and silver, respectively. We select from the following seven common dielectric materials: Al_2O_3, HfO_2, MgF_2, SiC, SiN, SiO_2, and TiO_2.[29−31] Since the substrate is made of silver, which light cannot transmit through, the emissivity/absorptivity $\epsilon(\lambda, \theta)$ is simply

$$\epsilon(\lambda, \theta) = 1 - R(\lambda, \theta) \tag{3}$$

where $R(\lambda, \theta) = \frac{1}{2}(R_{TE}(\lambda, \theta) + R_{TM}(\lambda, \theta))$ is the average reflectivity over the transverse electric (TE) and transverse magnetic (TM) polarizations. To obtain the angle-averaged emissivity $\epsilon_{avg}(\lambda)$, we perform the solid angle integral over a hemisphere:

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$$\epsilon_{avg}(\lambda) = \int d\Omega \cos\theta \epsilon(\lambda, \theta) \quad (4)$$

where $\int d\Omega = 2\pi \int_0^{\pi/2} d\theta \sin\theta$.

To find the optimum structure for radiative cooling given the objective spectrum specified above, we perform a broadband optimization on the emissivity spectrum. We choose to optimize in the normal direction for simplicity. For such a broadband device, we anticipate that while certain sharp spectral features generally exhibit a blue shift at oblique angles of incidence, overall its spectrum should not exhibit a strong angular dependence as it is inherently nonresonant.[7] In Figure 1b, we plot the target emissivity spectrum as well as the solar spectrum and the atmosphere transparency spectrum. We set up the optimization for a maximum of $K = 9$ layers of materials, and we first randomly generate a population size of $N = 1000$ with different materials that have thicknesses randomly and uniformly distributed between 0 and 2 μm. We heuristically set the reselection rate $r = 0.90$ and the mutation rate $\mu = 0.01$, and we perform local optimization for refinement every $G = 15$ generations for the top $P = 60$ individuals. Over the course of optimizing for 75 generations, in Figure 1d, we plot the best and average merit function in each generation while indicating the generations where refinement occurred. As expected, the evolutionary algorithm systematically produced structures that generally decreased the merit function from one generation to the next, and the optimization converged after 60 generations to a six-layer structure with the following material composition and thicknesses shown in Table 1.

Table 1. Material Composition and Thicknesses for the Optimized Radiative Cooler

layer #	material	thickness (μm)
	air	superstrate
1	MgF_2	3.279
2	HfO_2	0.640
3	SiC	0.657
4	MgF_2	1.940
5	SiC	1.337
6	MgF_2	0.047
	Ag	substrate

We now analyze the emissivity properties of the structure in Table 1. In Figure 1b, we plot the normal-direction emissivity with red dotted lines. The emissivity spectrum has the same feature as the target emissivity, including the suppression of absorption in both the solar spectrum and the ozone absorption line, as well as the enhancement of emissivity in the wavelengths in the range of 8 to 13 μm that is outside the ozone absorption line. We plot the emissivity spectra at different angles in Figure 1c. The emissivity in general shows only weak angular dependency as expected. In particular, the prominent dip in the emissivity spectrum around the wavelength of 9.5 μm, which is desirable, persists over a wide range of angles. The overall angle-averaged emissivity, as calculated according to eq 4, is shown as the blue solid line in Figure 1b. While we do not directly optimize for such angle-averaged emissivity, the resulting angle-averaged emissivity reproduces the main features of the target emissivity spectrum.

To assess the cooling performance of the device above, we compare the lowest temperature achievable between our device and the device in ref 7 under the same thermal insulation and solar illumination. From the emissivity spectra in Figure 1b, we find that this structure absorbs 3.5% of the incident sunlight (compared to 3% from ref 7), and it has a cooling power of 45.7 W/m^2 (compared to 38.0 W/m^2 from ref 7) at the ambient temperature. Assuming the absence of parasitic heat load (i.e., by setting the parasitic heat load coefficient $h = 0$ in ref 7), we find that the six-layer structure can achieve a steady-state temperature that is 23.4 K below ambient temperature, as compared with 19.5 K below ambient for the device in ref 7. This temperature reduction directly results from the reduction of emissivity between the 9.3 to 10 μm wavelength range. Furthermore, in Figure 1e, we compare the cooling power between our six-layer device and the one in ref 7 at a given temperature, and we find that our optimized structure has a higher cooling power than the device in ref 7 at all temperatures, which enables our device to more efficiently achieve lower temperatures. Therefore, our memetic algorithm has allowed us to find a multilayer structure that has better cooling performance, as evaluated by both metrics, using one fewer material layer.

OPTIMIZATION OF INCANDESCENT LIGHT BULB FILTER

We now demonstrate the use of the memetic algorithm in the optimization of thin-film filter for an incandescent light bulb. While incandescent light bulbs have attractive properties such as having a high color rendering index (CRI), they suffer from poor efficiencies of around 3% because most of their emission is lost as near-infrared radiation (which does not contribute to illumination in the visible wavelength range).[14] The efficiency of incandescent light bulbs, however, can be improved by enclosing them inside a spectral filter (shown schematically in Figure 2a), which has high transmittance in the visible wavelength range and high reflection in the near-infrared wavelength range.[14] By limiting infrared transmission, the reflected infrared light can be recycled to heat up the light bulb's emitter. The emitter can thus achieve a higher temperature at a lower input power, which is conducive to enhanced emission in the visible spectrum and therefore improved efficiency. This concept was demonstrated by Ilic et al., who fabricated a filter structure using 90 layers of alternating Ta_2O_5 and SiO_2 to surround a high-temperature tungsten emitter.[14] Ilic et al. experimentally demonstrated that such a 90-layer structure could improve the efficiency of incandescent bulbs from around 3% to 6.6%. However, from economic, fabrication, and structure stability considerations, it would be preferable to use fewer layers of materials.

Here, we implement the memetic algorithm to design an improved spectral filter for incandescent light bulbs. In particular, we show that the memetic algorithm can design a structure that achieves more efficient visible light emission as compared to the device in ref 14 while using fewer than half the number of material layers. Similar to the radiative cooler design in the previous section, we design for spectrum in the normal direction and then consider the angular properties of the resulting structure after optimizations. The target reflection spectrum in the normal direction is shown in Figure 2b, and we expect the broadband transmission window in the visible wavelengths to persist over a wide range of angle as was the case in ref 14. We optimize with the maximum number of layers being $K = 45$ and a population size of $N = 3000$, and the materials are selected from the same seven materials as is used in the previous section. The thickness for each layer is

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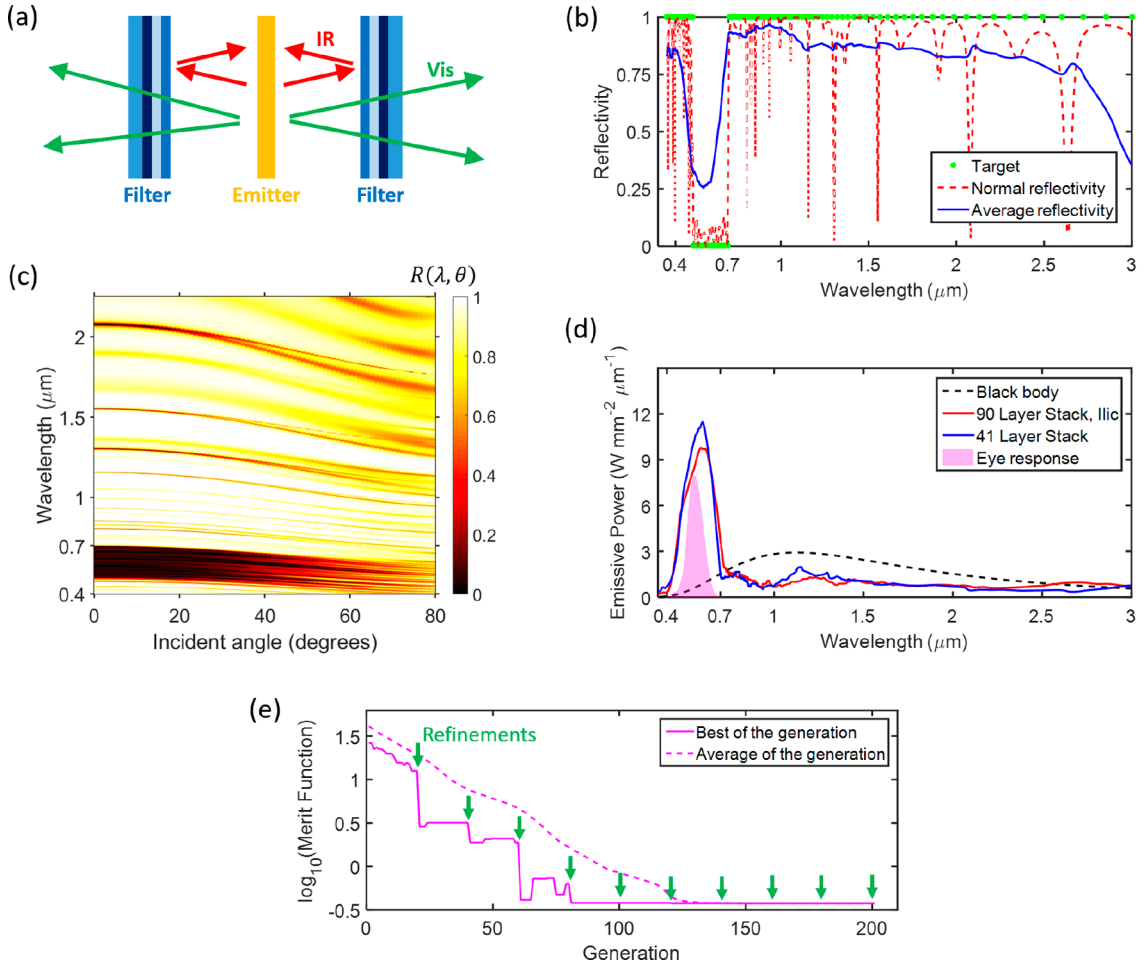


Figure 2. (a) Schematic of the incandescent light bulb filter. (b) The reflectivity spectrum of the designed incandescent filter. The green dots are the target reflectivity spectrum in the normal direction. The red dotted line and the blue solid line are the normal and angle-averaged reflectivity spectrum of the structure shown in Table 2, respectively. (c) Angle-dependence of the reflectivity spectrum for the structure shown in Table 2. (d) Comparison between the structure in Table 2 (blue) and ref 14 (red) given the same input power of 100 W to the emitter. The eye's responsivity and a blackbody's emissive power spectrum at the same input power are plotted for reference. (e) Progression of the light bulb optimization merit function for the best and average of the structures in each generation. The green arrows indicate the generations where the refinement steps took place.

Table 2. Material Composition and Thicknesses for the Incandescent Light Bulb Filter

layer #	material	thickness (nm)	layer #	material	thickness (nm)	layer #	material	thickness (nm)
	air	NA	15	HfO_2	158	30	MgF_2	170
1	MgF_2	120	16	SiN	190	31	SiO_2	217
2	SiC	124	17	MgF_2	213	32	SiC	106
3	Al_2O_3	242	18	SiC	123	33	MgF_2	345
4	SiO_2	166	19	MgF_2	223	34	Al_2O_3	244
5	SiC	227	20	HfO_2	109	35	TiO_2	229
6	MgF_2	241	21	SiO_2	181	36	Al_2O_3	243
7	SiC	26	22	TiO_2	19	37	SiO_2	139
8	MgF_2	238	23	Al_2O_3	191	38	SiC	111
9	SiC	221	24	SiC	244	39	MgF_2	185
10	MgF_2	193	25	TiO_2	75	40	SiC	105
11	SiC	92	26	MgF_2	199	41	MgF_2	312
12	MgF_2	212	27	SiC	198		air	NA
13	SiC	128	28	SiO_2	172			
14	MgF_2	222	29	HfO_2	246			

randomly initialized between 0 and 300 nm. The reselection and mutation rates are set to be r = 0.90 and μ = 0.01, respectively, and we perform local optimization for refinement every G = 20 generations for the top P = 30 individuals. Over the course of 200 generations of optimization, in Figure 2e, we plot the best and average merit function in each generation and

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indicate the generations where refinement occurred. The algorithm converged in 132 generations to a 41-layer structure, whose parameters are listed in Table 2:

We now analyze the optical properties of the structure described in Table 2. In Figure 2b, we plot the structure's normal-direction reflectivity, which reproduces excellently the main requirements of the target spectrum. We further calculate the structure's angular dependence and plot it in Figure 2(c). With this, we can compute the average reflection across all angles, which is plotted as the blue solid line in Figure 2(b). As desired, the angle-averaged reflection is low in the visible range to allow high visible light transmission between 400 and 700 nm, but the reflection is high in the infrared part so that infrared radiation can be recycled to further heat up the emitter.

To directly evaluate the structure in Table 2 against the structure in ref 14., we surround incandescent emitters with the respective structures and compare the enhancement factor of visible light emission at a fixed input power. For simplicity, we assume that the emitter is a perfect blackbody, and the multilayer filter encloses the emitter with a view factor of f = 0.95, as was the case in ref 14. In this setup, the total power emission from the emitter can be calculated as

$$P_{\text{emitter}}(T) = A \int \mathrm{d}\lambda I_{\text{emitter}}(\lambda, T)\epsilon_{\text{eff}}(\lambda) \tag{5}$$

where A is the emitter surface area, $\epsilon_{\text{eff}}(\lambda)$ is the angle-averaged effective emissivity computed from eq 4 using the view-factor adjusted emissivity[14] $\epsilon'(\lambda,\theta) = 1 - f^2R(\lambda,\theta)$, and $I_{\text{emitter}}(\lambda)$ is the emission intensity spectrum of the blackbody emitter described by Planck's law:

$$I_{\text{emitter}}(\lambda, T) = \frac{2hc^2}{\lambda^5}\frac{1}{e^{hc/(\lambda k_B T)} - 1} \tag{6}$$

To measure the enhancement of visible light emission due to the spectral filter, we compute the emissivity enhancement factor χ as[14]

$$\chi = \frac{\int \mathrm{d}\lambda \epsilon_{\text{eff}}(\lambda) I_{\text{emitter}}(\lambda) V(\lambda)}{\int \mathrm{d}\lambda I_{\text{emitter}}(\lambda) V(\lambda)} \tag{7}$$

where $I_{\text{emitter}}(\lambda)$ is the intensity emission spectrum of the blackbody emitter, and $V(\lambda)$ denotes the eye's sensitivity spectrum[32] and is plotted in Figure 2d. Note that χ = 1 when there is no spectral filtering, and χ > 1 for filters that favor emission in visible wavelengths over infrared and ultraviolet wavelengths.

We can now numerically compare the performance of our stack structure with the structure in ref 14. First, we fix the input power to the emitter to be P_{emitter} = 100 W, and we assume that the surface area of the emitter is A = 20 mm^2, as is reasonable for an incandescent light bulb.[14] We assume that the heat in the emitter is entirely dissipated through radiation, i.e. there is no conductive or convective heat loss. With the angle-averaged emissivity spectrum of our structure and the structure in ref 14 (computed using the information supplied in the article), we can then solve for the temperature in eq 5 to find that our structure reaches a temperature of T_1 = 3743 K, whereas the structure from ref 14 reaches a temperature of T_2 = 3599 K. For comparison, at the same input power of 100 W, a blackbody would reach a temperature of T_{BB} = 2577 K. The overall thermal emission intensity spectrum of our structure and Ilic et al.'s structure are plotted along with the eye's sensitivity spectrum[14,32] in Figure 2d, where we see that the filtered emitters have a highly enhanced emission in the visible range as compared with a blackbody emitter at the same input power. We then calculate χ in eq 7 for both structures and find that our 41-layer stack achieved χ_1 = 14.8, compared to Ilic et al.'s 90-layer structure that achieves χ_2 = 12.8. Therefore, our 41-layer stack achieves a larger enhancement factor for visible-light emission than the 90-layer structure in ref 14. Moreover, the overall thickness of the 41-layer stack is 7.4 μm as compared to 15.8 μm for the structure in ref 14, and such thickness reduction is beneficial from an economic and fabrication perspective. Through this example, we have demonstrated that the memetic algorithm allows us to design an incandescent light bulb filter with improved performance, while using fewer than half the number of layers compared to the previous design.

DISCUSSION AND SUMMARY

In both of the memetic algorithm demonstrations above, we have heuristically chosen the reselection rate to be r = 0.90 and mutation rate to be μ = 0.01. On the other hand, it is well-known that in evolutionary algorithms, the choices of r and μ is important and can influence the performance of the algorithm depending on the properties of the search landscape. In part 1 of the Supporting Information, we provide a study that examines the effects of using r = [0.80,0.90,0.99] and μ = [0.01,0.05,0.10], as well as fixing r = 0.90 and having very high mutation rates of μ = [0.50,0.80]. These examples both show that a high reselection rate generally accelerates the decrease in the merit function during early generations. As for the end-result of these optimizations, while a moderate reselection (r = 0.90) and high mutation rate (μ = 0.10) are beneficial for the radiative cooling device optimization, a low reselection rate (r = 0.80) and high mutation rate (μ = 0.10) more consistently generate higher-performing structures in the light bulb filter optimization. In addition, having very high mutation rates of μ = 0.50 and 0.80 could help the memetic algorithm converge to better-performing structures for both optimizations. This study illustrates that an optimal choice of the reselection and mutation rates is problem-specific, and it is important to diversify the population in order to explore more regions of a search landscape.

From both the radiative cooling device and incandescent light bulb filter examples above, the resulting structures as obtained from our memetic algorithm have higher performance than those reported in the literature while using fewer number of layers. In these examples, we use a larger set of materials as compared to refs 7 and 14. As an additional comparison with the multilayer structures used in refs 7 and 14, we apply this memetic algorithm to optimize the layer thicknesses of seven layers of SiO_2 and HfO_2 for the radiative cooling device and 90 layers of SiO_2 and Ta_2O_5 for the incandescent light bulb filter. The details of these optimization results are provided in part 2 of the Supporting Information, where we show that the memetic algorithm produced a seven-layer radiative cooling device that could reach an equilibrium temperature of 20.8 K below an ambient temperature of 293 K, and an 80-layer incandescent light bulb filter with an emissivity enhancement factor of χ = 14.1. Both of these devices could out-perform those reported in the literature, but they have lower performance compared to the structures in Tables 1 and 2, which were optimized using a more diverse set of materials. These two examples demonstrate that the memetic algorithm could more effectively perform optimizations with the same

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materials that were used in refs 7 and 14. Furthermore, the results suggest that when designing spectral properties of a structure, it is advantageous to build the structure from a diverse set of materials.

In summary, we have presented an implementation of a memetic algorithm for the design of multilayer optical thin-films that is especially suited for broadband thermal and energy applications. Given a target optical spectrum, we construct the optimization as a mixed integer problem, where the optimization variables consist both a list of discrete variables that represent different materials and a list of continuous variables that represent each material's thickness. As a set of demonstrations, we have employed the memetic algorithm to optimize the performances of both a passive radiative cooling device and an incandescent light bulb filter. In both examples, by using a diverse set of materials, the multilayer structures obtained from the optimization have fewer layers but better performances as compared to those presented in the literature. Because of its versatility and effectiveness, our algorithm should prove to be a powerful tool in the optimization of multilayer optical materials, not only for thermal and energy applications, but also for imaging, display, and information processing applications.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsphotonics.7b01136.

Figures on the study of various reselection rates r and mutation rates μ on the performance of the memetic algorithm; figures on the properties of the optimized radiative cooling device and incandescent light bulb filter using just two materials; tables of the layer thicknesses for the optimized radiative cooling device and incandescent light bulb filter using just two materials (PDF).

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